

Albin John

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EDUCATION

Indian Institute of Information Technology, Kottayam (IIIT Kottayam)

2023 – 2027

B.Tech in Computer Science and Engineering, Specialization in AI & Data Science

CGPA: 8.94/10

EXPERIENCE

AI Intern

Feb 2026 – Present

Klyonix Tech Consulting

- Designed a **multi-tenant chatbot** using a **RAG + Text-to-SQL architecture** for natural-language querying of relational databases and real-time business insights
- Built a dynamic SQL generation pipeline to extract key metrics, **reducing reporting effort by 60%** and improving accuracy on temporal (date/time-based) queries
- Implemented **role-based access control (RBAC) guardrails at the LLM layer** to enforce secure, hierarchy-based data access while maintaining high accuracy in natural language to SQL translation

AI & Computer Vision Intern

May 2025 – Aug 2025

AI4SEES

- Built an end-to-end computer vision system on **YOLOv8** for real-time object detection, segmentation, and hazard tracking, then fine-tuned a theft-detection model on tricky edge-case footage
- Optimized the inference pipeline to keep pace with high-FPS video and wired the outputs into a **React.js** dashboard for live monitoring

PROJECTS

Casper – Task-Oriented Autonomous AI Agent | [GitHub](#)

Feb. 2026

Python, LangGraph, AutoGPT, Tool-Calling, FastAPI, React.js

- Built an autonomous agent in **LangGraph** that reasons over multiple steps and reflects on its own output, calling external APIs to get real-world tasks actually done
- Added a state layer so the agent keeps context across long, multi-turn workflows instead of starting fresh each time served through **FastAPI**, with a **React.js** UI for watching it work
- Capped agent loops by disabling tool access once the search budget runs out, and streamed live reasoning to the frontend via Server-Sent Events.

LegalEase – Advanced RAG-based Legal Advisor | [Demo](#) | [GitHub](#)

Oct. 2025

Python, ChromaDB, FlashRank, RAGAS, OpenAI API, Streamlit

- Built a multi-stage **RAG** pipeline with semantic chunking, a **ChromaDB** vector store, dense retrieval, and **FlashRank** re-ranking, so every answer stays grounded in the actual source text
- Set up an automated evaluation pipeline with **RAGAS** to track answer quality, hitting **92% faithfulness** and **88% answer relevance** with real numbers behind the hallucination reduction
- Improved retrieval precision with a two-stage retrieve-then-rerank pipeline (top-25 narrowed to top-8 via cross-encoder) and added a refusal guardrail so the model won't answer or invent citations when nothing relevant is retrieved.

SpiceNest – Full-Stack E-Commerce Platform (Live in Production) | [Live Site](#) | [GitHub](#)

May 2026

React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, Tailwind CSS (v4), Razorpay SDK, JWT

- Built and shipped a full-stack e-commerce platform – **React.js** and **Tailwind CSS v4** on the frontend, a **Node.js/Express.js** REST API on the backend for smooth product discovery and browsing
- Modeled a secure relational schema in **Prisma** on Neon's serverless PostgreSQL, with **bcryptjs** for password hashing and **JWT** for authentication
- Hooked up **Razorpay** for secure checkout and used the **React Context API** to keep cart state in sync across the UI in real time

Influential Node Identification using Graph Convolutional Networks 2026 – Present

B.Tech Honors Project, IIIT Kottayam — Python, PyTorch, Graph Neural Networks, NetworkX

- Developing a **GCN** model to spot the most influential spreader nodes in a network – useful for understanding information diffusion and epidemic control
- Generated ground-truth influence labels by running **SIR epidemic simulations** on each network and ranking nodes with the discrimination-index method
- Fixed a train/test leakage flaw common in prior work by training on synthetic Barabási–Albert graphs and testing on unseen real-world networks, then benchmarked against InfGCN, DeepInfNode, NLGCN, and CGCAN.

Development Aware Flood Susceptibility Rating using Satellite Data 2026 – Present

B.Tech Final Year Project, IIIT Kottayam — Remote Sensing, Geospatial ML

- Building a flood susceptibility model that uses **multi-temporal satellite data** to track land-use change over time, instead of relying on a single static terrain snapshot
- Factoring built-up growth and land-cover change directly into the susceptibility score, so the rating reflects how urbanization actually shifts flood risk

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL (PostgreSQL/MySQL), C/C++, Java, HTML/CSS

LLM & Agentic AI: RAG Pipelines, LangChain, LangGraph, Multi-Step Agents, Tool-Calling, Text-to-SQL, Prompt Engineering, Embeddings, Re-ranking, Vector Databases (ChromaDB)

AI Evaluation & Guardrails: RAGAS (Faithfulness / Answer Relevance), Hallucination Reduction, LLM-layer RBAC

AI/ML & Computer Vision: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLOv8, Graph Neural Networks, NumPy, SciPy

Web & Backend: FastAPI, Flask, Django, Node.js, Express.js, REST APIs, React.js, Vite, Tailwind CSS, Streamlit

Databases & DevOps: PostgreSQL, Prisma ORM, Neon, Git, GitHub, Docker, Vercel, n8n

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

ACHIEVEMENTS

- Solved **400+ problems on LeetCode**, strengthening problem-solving in data structures and algorithms
- **GATE Qualified** (Computer Science & Information Technology)
- Secured **97.5 percentile in JEE Main**
- Ranked **under 500** in state-level engineering entrance examinations including **KEAM** and **CUSAT CAT**
- **Finalist in multiple hackathons**, building AI-driven prototypes under tight deadlines
- Awarded the **Rajya Puraskar** (Bharat Scouts & Guides, state-level award, 2018) and the **Nanmamudra** by the Govt. of Kerala for service in the Chief Minister's Shield Competition (2022)

CERTIFICATIONS

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| • Supervised Machine Learning: Regression and Classification – Stanford & DeepLearning.AI | May 2025 |
| • The Complete Full-Stack Web Development Bootcamp (61.5 hrs) – Udemy | Feb 2025 |
| • Create Your Own Chatbot Website with Open Source LLMs – IBM (Cognitive Class) | Jul 2024 |
| • Prompt Engineering for Everyone – IBM (Cognitive Class) | Jun 2024 |
| • Infosys Springboard: Intro to NLP, MERN Stack & FSD, Information Security | Nov 2024 |
| • Python Libraries for Data Science – Simplilearn SkillUP | Jun 2024 |

LEADERSHIP & EXTRACURRICULAR

House Captain & Badminton Sub-Lead Oct. 2025 – Present

Sportec – IIIT Kottayam

- Led a contingent of **500+ students** as House Captain, coordinating **15+ events** and securing **Runner-Up** in the intra-college sports meet
- Represented IIIT Kottayam in **National-level Inter-IIIT** badminton tournaments, demonstrating competitive performance, teamwork, and leadership

Social Media Lead Aug. 2025 – Aug. 2026

Cultural Club – IIIT Kottayam

- Managed the club's college social media pages, planning and posting content to promote events and boost student engagement
- Edited and uploaded visual and video content for campaigns, event promotions, and recaps